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DICTIONARY TO GENEBASE 'DICOTOBASE' USER GUIDE

SKILL	PROJECT NAME	PLM
Software	AGL	

ABSTRACT / CONCLUSION

This document is a user guide for DicoToBase tool.

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[See Signature Workflow Instruction](#)

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1 Revision Summary

Version	Date	Modif N°	Author	Modified paragraphs and kind of modification
01	24/02/11	01	Amr Abdelrahman	Creation
02	14/02/2012	01	Mahmoud Sameh	Updated For DicoToBasev1.2 the new argument /NoMultiGroup added.
02	22/03/2012	02	Mohamed Safwat	Added a usage note about the argument /NoMultiGroup
02	12/4/2012	03	A.Abdelrahman	Added new Section 4.4 for compatibility of generated files with BaseDev application.

2 Subject matter of the document

This document provides a quick and fast information on how to use DicoToBase tool and also shows the expected differences between the generated files from the tool and the generated from access M65 BaseDev.

3 Reference documents

- V01 NT 10 02934 01 – USER REQUIREMENTS FOR A DICTIONARY TO GENEBASE CONVERTER.

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4 DicoToBase Tool

4.1 Introduction

DicoToBase is a command line tool that was made to replace projet "Generate Genebase and Winades txt files" step as projet was always very slow and also many times has unexpected crashes, so the tool input will be the project dictionary txt files (donnee.txt, refcdc.txt, typedef.txt, cfgbdd.txt); fonction.txt and requirements.txt are not needed for DicoToBase current version.

The output files of the tool are the txt files that will be used as input for Genebase later (caracter3.txt, caractab3.txt, comment3.txt, groupes3.txt, enumere3.txt, referenc3.txt) and (caracter.txt, caractab.txt, comment.txt, groupes.txt, enumere.txt, referenc.txt).

Note:

***2.txt files that is used for genebase to generate BD2 files are not generated and not supported by DicoToBase tool current version. Therefore the minimal version of Genebase which can be used after this tool is the version S_50 .**

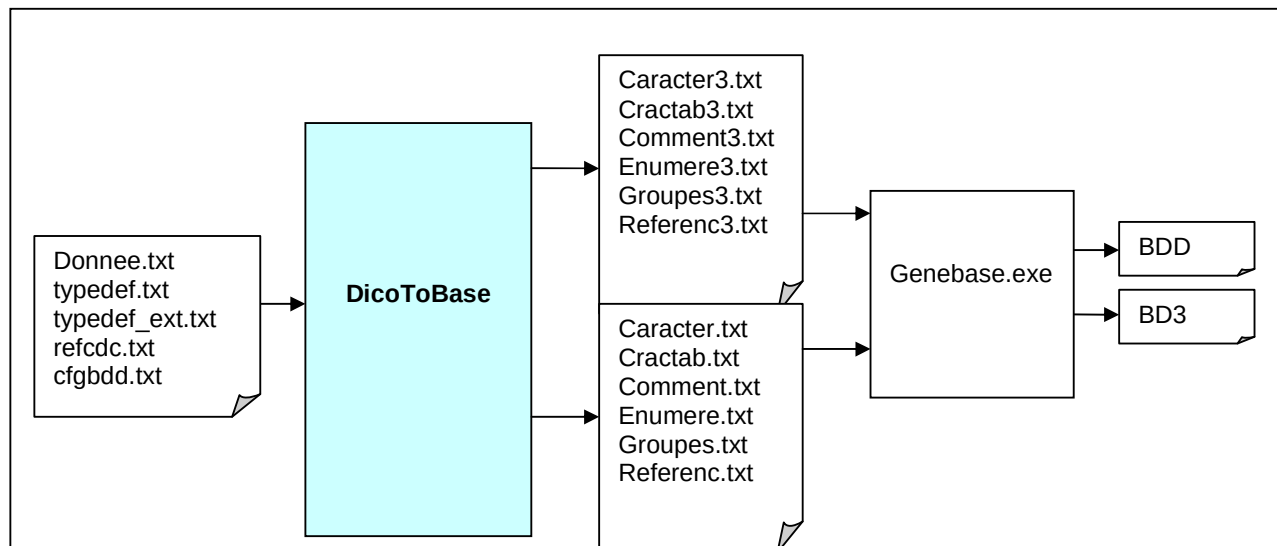


Figure 1: DicoToBase Tool Input and Output Files

4.2 Optional input file : "typedef_ext.txt"

Access 97 makes no difference between data types which differ only in case ; for example, "UINT32" is the same as "UInt32" and the same as "uint32". Therefore the "typedef.txt" file can only contain one type definition, typically "uint32" .

However, some projects require types like "UInt32" which only differ in case.

DicoToBase is stricter than Access when it looks for types definitions, and searches for exact matches. Therefore an additional text file is taken as input of the tool, "typedef_ext.txt" . This file has the same format as the typedef.txt .

DicoToBase comes with a default "typedef_ext.txt" file, which is copied into the input directory if the file does not already exist.

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4.3 How To Use

DicoToBase takes 4 optional arguments:

DicoToBase.exe [/inDir:"input files directory"] [/outDir:"output files directory"] [/mysqlDir:"temp dir"] [/NoMultiGroup]

/inDir	Input files directory path; if not set the tool will assume that the input files are in current directory
/outDir	Output files directory path; if not set the tool will generate the output files in the current directory.
/mysqlDir	A directory where the tool will create a subfolder 'MySQLdata' to store temporary tables
/NoMultiGroup	When used, it Disable the Export of Groupe_1 and Groupe_2 for each variable/parameter in the generated groupe.txt and groupe3.txt files. This option MUST be used for Genebase versions prior to 5.9.

Use case examples:

1. If the input files into c:\Test directory and we want to generate the output files in the same directory use:
C:\Test>startenv dicotobase
2. if the input files are in c:\Test\input and we want the output files to be in c:\Test\Output use:
C:\Test>startenv dicotobase /inDir:.input /outDir:.output
3. To store temporary files on another drive, for example in folder "b:\MySQLdata", use :
C:\Test>startenv dicotobase /mysqlDir=b:

Folders with spaces can be surrounded by double quotes, for example :

C:\Test>startenv dicotobase /inDir:"e:\Task 225\src"

Arguments are not case sensitive, and the symbol "=" or ":" can be used indifferently to separate options and folder names.

4.4 Tool reports

The application generates two reports in the output folder:

DicoToBase.log : contains errors and warnings specific to the tool, for example missing types in the TYPEDEF.TXT file.

BD3_TO_BDD.log : list the items which have been truncated in the "caracter.txt" file due to database limitations, for example long parameter names truncated to 32 characters; this report is similar to the report generated by tool "BDconverter"

4.5 Output compatibility with BaseDev Generated files,

DicoToBase	BaseDev	Comment
1.0	M65	DicoToBase 1.0 Generated files are compatible with generated files from BaseDev M65
1.1	M65	DicoToBase 1.0 Generated files are compatible with generated files from BaseDev M65
1.2	M68	DicoToBase 1.2 Generated files are compatible with generated files from BaseDev M68. It is also compatible with generated files from M65 when the option

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		/NoMultiGroups is used, this option is mandatory to generate files that will be used with GeneBase Version prior to 5.9.
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4.6 Differences Justifications

This part of the document is not mandatory to know how to use the tool, but it is mandatory if someone wants to validate the output files of the tool with the output files generated from BaseDev Projet Tool.

There are some differences between the output files from DicoToBase tool and the output files generated from BaseDev Projet access tool; most differences come from :

- different identifiers because data are ordered differently.
- the empty default value " does not exist in the BD3, it is replaced by a null value (0).

Note:

All justifications made for *.txt (BD3) files apply to *.txt (BDD) files because the files used for BDD generated come from internal BDConverter.

File	Caracter.txt
Difference description	Alias2 can be different
Output DicoToBase	2 Abs_mux Abs_mux Abs_mux Abs_mux 0000 0 1 0 0 1 0 bool 2 0.0 0.0 0 1
Output Projet M65	2 Abs_mux Abs_mux Abs_mux 0000 0 1 0 0 1 0 bool 2 0 0 1
Justification	Alias2 are not managed in Caracter3, so they are copied from Alias1 value during internal conversion.
Expected Effect	"Alias 2" parameter information may be wrong in Winades

File	Caracter3.txt
Difference description	Records Id's are always different.
Output DicoToBase	1 IGN_HAL_MODE_ADAP Ign_hal_mode_adap IgnHALMode 0000 0 1 2 bool 0 0 0 0 1 0 2 0 0 0 IGNHAL 0 1 3 3 0 0 0 0 0 0 0 0 0 0 0 0 IGN_HAL_MODE_FIXED Ign_hal_mode_fixed IgnHALMo
Output Projet M65	5 IGN_HAL_MODE_ADAP Ign_hal_mode_adap IgnHALMode 0000 0 1 2 bool 0 0 0 0 1 0 2 0 0 0 IGNHAL 0 1 3 3 0 0 0 0 0 0 0 0 0 0 0 0 IGN_HAL_MODE_FIXED Ign_hal_mode_fixed IgnHALMo
Justification	Records Id's are always different between files exported from DicoToBase and the files exported from Projet. It is normal as ids exported from projet are auto generated and do not always start with 1 as for DicoToBase.
Expected Effect	None

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File	Caracter3.txt
Difference description	Sometimes (in TUE5 project) the offset is a negative number. It is exported from Projet as a negative number and from DicoToBase as (2^32 + negative number).
Output DicoToBase	P_actionneurs_cc_eep_count P_actionneurs_cc_eep_count Panne_fonc_type_eep_count4 0000 4294967201 1 0 - 0 0 0 0 1 3 0 EEPROM 0 1 3 3 0 0 0 0 0 1 0 0 0 1
Output Projet M65	P_actionneurs_cc_eep_count P_actionneurs_cc_eep_count Panne_fonc_type_eep_count4 0000 -95 1 0 - 0 0 0 0 1 3 0 EEPROM 0 1 3 3 0 0 0 0 0 1 0 0 0 1
Justification	BD3 do not support negative offset numbers as offset is "4 unsigned bytes" so the negative number will have its unsigned representation. This will not affect tools like Genebase or Transfor because the physical parameter address will remain the same, due to the assumption that microcontroller address space wraps at 2^32 .
Expected Effect	None

File	Caracter3.txt
Difference description	Some times Min & Max Decimal values has non integer values into caracter3 exported from projet. They are converted to integer into caracter3 exported from DicoToBase
Output DicoToBase	VfFPPR_phi_PmpLen VfFPPR_phi_PmpLen VfFPPR_phi_PmpLen 0000 0 4 0 - 0 0 0 2 1 0 0 A2L AUT RTWEC 0 0 3 3 0 1023 0 1023.98 0 1 0 0 0 1
Output Projet M65	VfFPPR_phi_PmpLen VfFPPR_phi_PmpLen VfFPPR_phi_PmpLen 0000 0 4 0 - 0 0 0 2 1 0 0 A2L AUT RTWEC 0 0 3 3 0 1023 0 1023.98 0 1 0 0 0 1
Justification	The internal library holds Min and Max decimal values into signed integer so it can't hold non integer values; this is justified since decimal values are always expected to be integer values, not floating point values (which is a defect in Temp/Projet).
Expected Effect	None

File	Caracter3.txt
Difference description	DicoToBase calculates the bounds "min/max phy/dec" from each other even if the transfer function was empty as it considers empty transfer function the same as identity (0).
Output DicoToBase	GDU_noThdScg_PwrRly_C GDU_noThdScg_PwrRly_C GDU_au16NoThd[57] 0000 0 2 0 - 0 0 0 0 2 1 0 GD 1 1 3 3 0 32767 0 32767 0 1 0 0 0 1
Output Projet M65	GDU_noThdScg_PwrRly_C GDU_noThdScg_PwrRly_C GDU_au16NoThd[57] 0000 0 2 0 - 0 0 0 0 2 1 0 GD 1 1 3 3 0 32767 0 0 0 1 0 0 0 1
Justification	Empty transfer function is the same as "0" (Identity) so DicoToBase sets the physical value equal to decimal ones if decimal ones are given, while projet not.
Expected Effect	None

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File	Caractab3.txt
Difference description	Records Id's is always different, the files are identical except the ids.
Output DicoToBase	2666 12 0 23210 0 5104 12 0 25166 0 5108 0
Output Projet M65	2050 12 0 23454 0 4545 12 0 20544 0 3434 0
Justification	The ids of the records are different but the links point to the same parameters in caractab3 file.
Expected Effect	None

File	Referenc3.txt
Difference description	Records Id's are always different, the files are identical except the ids.
Output DicoToBase	57 V03 NT 08 03070.5 AL_ALT_DGN: Diagnosis of Basic alternator load from sensor
Output Projet M65	60 V03 NT 08 03070.5 AL_ALT_DGN: Diagnosis of Basic alternator load from sensor
Justification	Records Id's are always different but the records have the same data and point to the same parameters in caractab3 file.
Expected Effect	None

File	Enumere3.txt
Difference description	Records Id's are always different, the files are identical except the ids.
Output DicoToBase	6 MCWDG_CHECKSUM_STATE 8 0 0 6 MCWDG_DATA_STATE 4 0 0 6 MCWDG_FAULT_STATE 16 0 0 6 MCWDG_IDENTIFIER_STATE 2 0 0 6 MCWDG_SYNCHRONOUS_STATE 1 0 0
Output Projet M65	7 MCWDG_CHECKSUM_STATE 8 0 0 7 MCWDG_DATA_STATE 4 0 0 7 MCWDG_FAULT_STATE 16 0 0 7 MCWDG_IDENTIFIER_STATE 2 0 0 7 MCWDG_SYNCHRONOUS_STATE 1 0 0
Justification	Records Id's are always different but the records have the same data, and the links used into caractab3 point to the same enumere3 records.
Expected Effect	None

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File	Comment3.txt
Difference description	Records Id's is always different.
Output DicoToBase	44 Generic received CAN frame 1
Output Projet M65	50 Generic received CAN frame 1
Justification	Records Id's are always different but the records have the same data and the links to caracter3 records are valid.
Expected Effect	No expected bad effect for this issue

File	Comment3.txt
Difference description	Comment3 generated from projet M65 has more records than the one generated from DicoToBase.
Output DicoToBase	
Output Projet M65	
Justification	Comment3 generated from Projet M65 in not coherent with caracter3 file, because it contains records with ids which do not exist in caracter3 file. The comment3 file generated from DicoToBase is more coherent with caractar3 file and doesn't contain extra records for non visualized winades records into dictionary.
Expected Effect	None

File	Groupes3.txt
Difference description	Records Id's are always different.
Output DicoToBase	0 1 0 APLS
Output Projet M65	0 3 0 APLS
Justification	Records Id's are always different but the links point to the same parameters.
Expected Effect	None

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File	Groupe3.txt
Difference description	Groupe3 generated from projet M65 has more records than the one generated from DicoToBase.
Output DicoToBase	
Output Projet M65	
Justification	Groupe3 generated from Projet M65 is not coherent with character3 file; it contains more records than character3 file and contains records with ids which is not found into character3 file. The groupe3 file generated from DicoToBase is more coherent than with character3 file and doesn't contain extra records for non visualized winades records into dictionary.
Expected Effect	None

File	Groupe3.txt
Difference description	The order of the records of groupe3 exported from DicoToBase is not exactly the same as the one exported from projet.
Output DicoToBase	0 1 0 APLS 0 2 0 Acquisitions 0 3 0 Add 0 4 0 AsAerVvt_Expclc 0 5 0 AsAerVvt_Expxxx 0 6 0 AsAerVvt_Inspclc
Output Projet M65	0 1 0 Acquisitions 0 2 0 Add 0 3 0 APLS 0 4 0 AsAerVvt_Expclc 0 5 0 AsAerVvt_Expxxx 0 6 0 AsAerVvt_Inspclc
Justification	The cause is that string comparison behavior differs between Access and MySQL in handling character case ('P' < 'c' < 'p', because ASCII codes are used during the comparison, whereas Access does a case-insensitive comparison) Lines order difference has no effect on the other tools later like Genebase.
Expected Effect	None

File	Groupe3.txt
Difference description	More groups can be created with DicoToBase than with projet.
Output DicoToBase	0 65 0 Divers 0 66 0 divers
Output Projet M65	0 65 0 Divers
Justification	Access is not case sensitive when dealing with text fields. Therefore with Access the groups "Divers" and "divers" are considered equivalent and all the data are stored into the same group. With DicoToBase, all treatments are case sensitive, so all the different groups will be created.
Expected Effect	More groups are created in the final A2L, and data are dispatched differently